CVPR JUNE 3-7, 2026 DENVER COLORADO

Project page

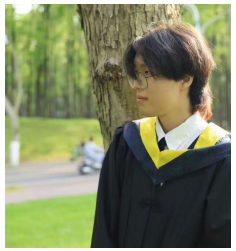
Plant Taxonomy Meets Plant Counting

A Fine-Grained, Taxonomic Dataset for Counting Hundreds of Plant Species

Huazhong University of Science and Technology, China

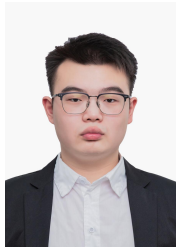

Jinyu Xu


Tianqi Hu


Xiaonan Hu

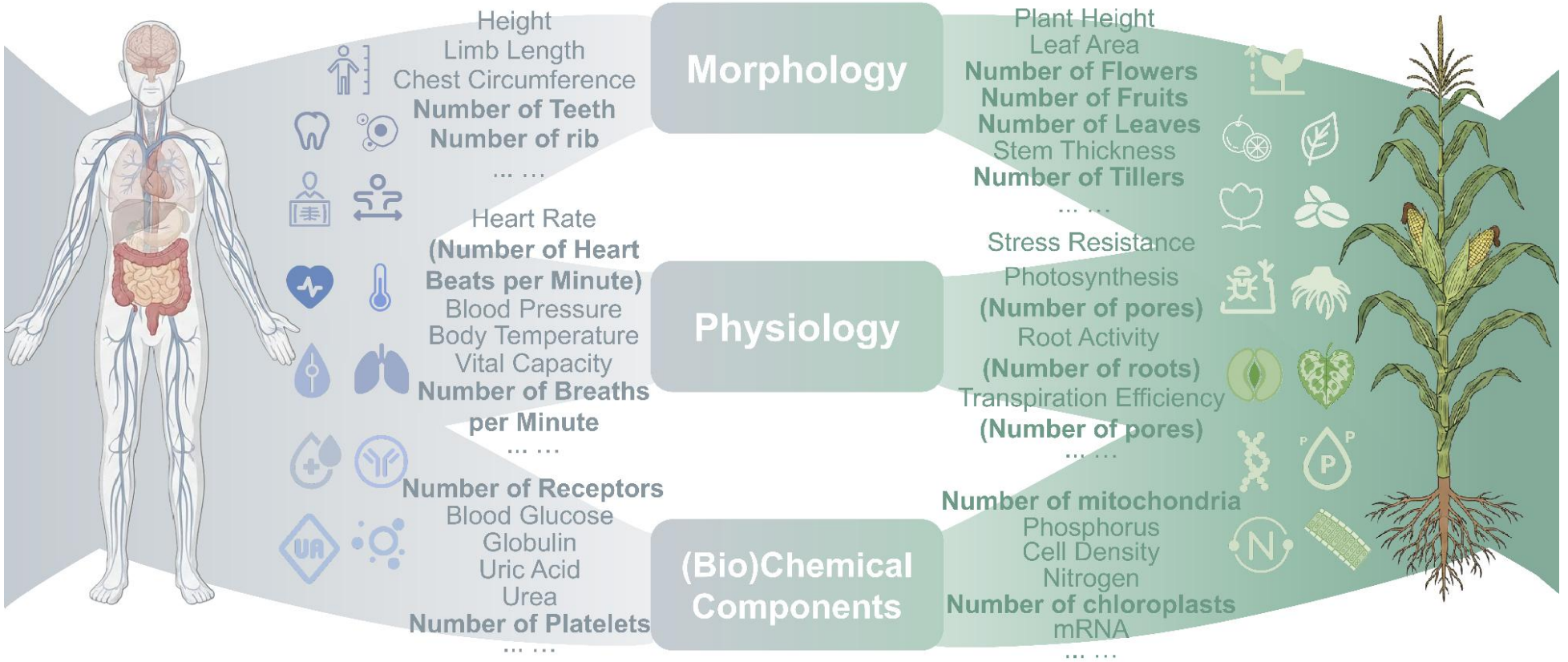

Letian Zhou


Songliang Cao


Meng Zhang


Hao Lu*

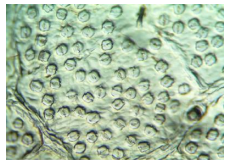
Plant Counting Is More Than “How Many”




Number of Flower


Number of Fruit


Number of Root

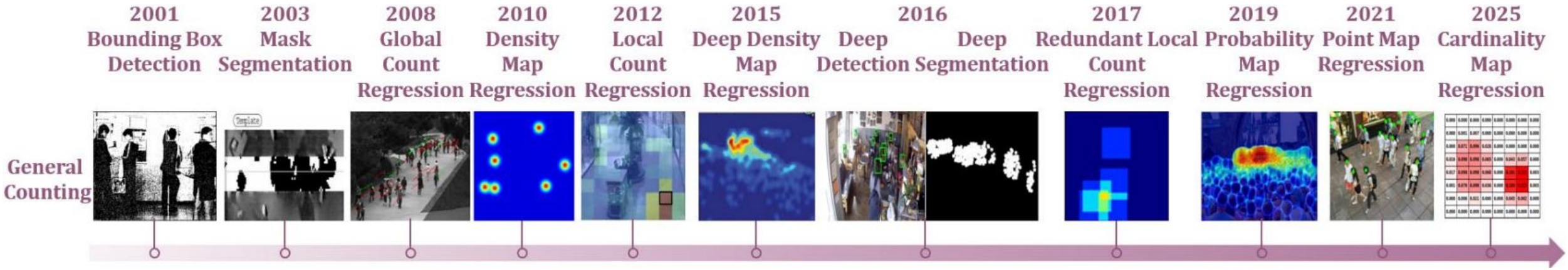

Number of Stomata


Number of Whole Plant


Number of Canopy

Akin to human health checks, plant counts are important indicators quantifying plant morphology, physiology, and chemical components.

Visual Counting: A Long-Standing Vision Problem



Evolution of visual counting paradigms



Crowds



Vehicles

Crowd and vehicle counting benchmarks are the main testbeds for visual counting.

Since 2008, visual counting paradigms have rapidly evolved.
But " what we count " has changed much less, mostly centered on crowds.

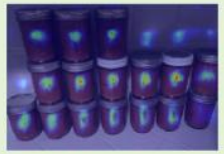
What We Count Shapes How We Count

Objects We Count

Crowds / Cars / Cells / Everyday Objects / ...

Methods We Design

Density-based



Detection-based



Prompt-based



Bias We Inherit: Hidden assumptions

Broad Labels
Counting often starts from simple and single classes.

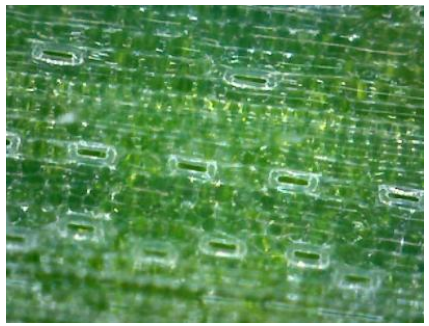
Repeated Targets
Many instances in one scene look alike.

Clear Instances
Each object is usually separable and countable as one unit.

Limited Scales
Most counting benchmarks stay within one visual scale.

What happens when the objects become "plants" ?











Plant Counting vs. Conventional Counting



Biodiversity vs **Homogeneity**





Fine Granularity vs **Coarse Category**





Nonrigid Structure vs **Rigid Shape**

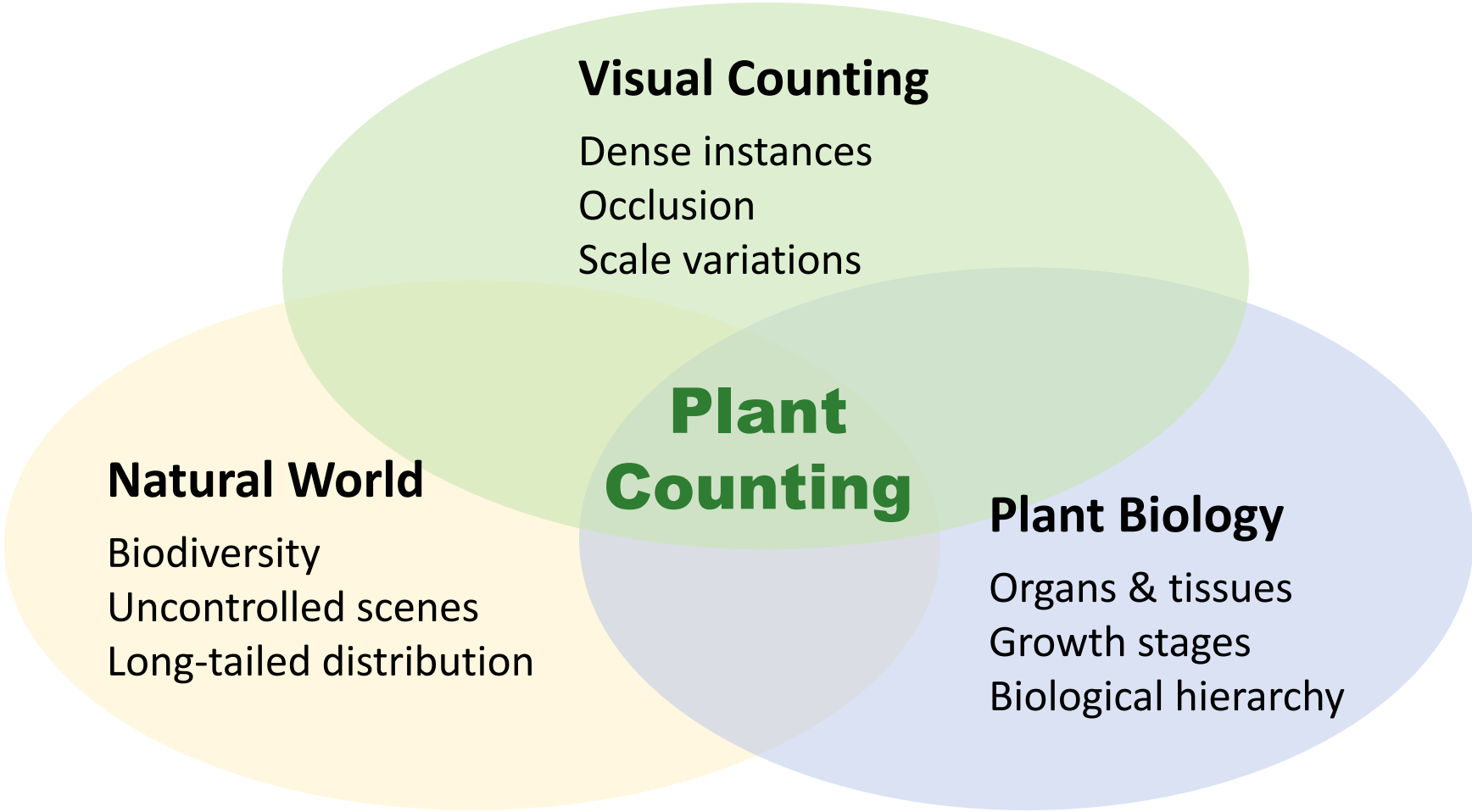




Spatiotemporal Variability vs **Physical Stability**



From Crowd Counting to Plant Counting



Plant counting invites visual counting into the natural world, combining biodiversity, biological structure, and multi-scale observation.

TPC–268: A Taxonomic Benchmark for Plant Counting

10,000
Images

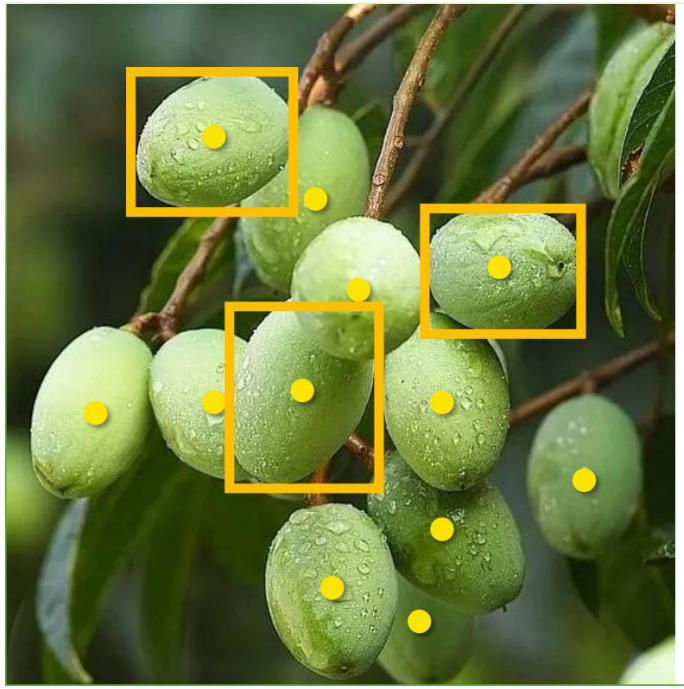
Scales from
Microscopy to UAV

678k+
Points Annotaitons

30,000
Bounding Boxes

242
Species
(Plantae & Fungi)

268
Countable Categories



Instances:
12 points

Exemplars:
3 boxes

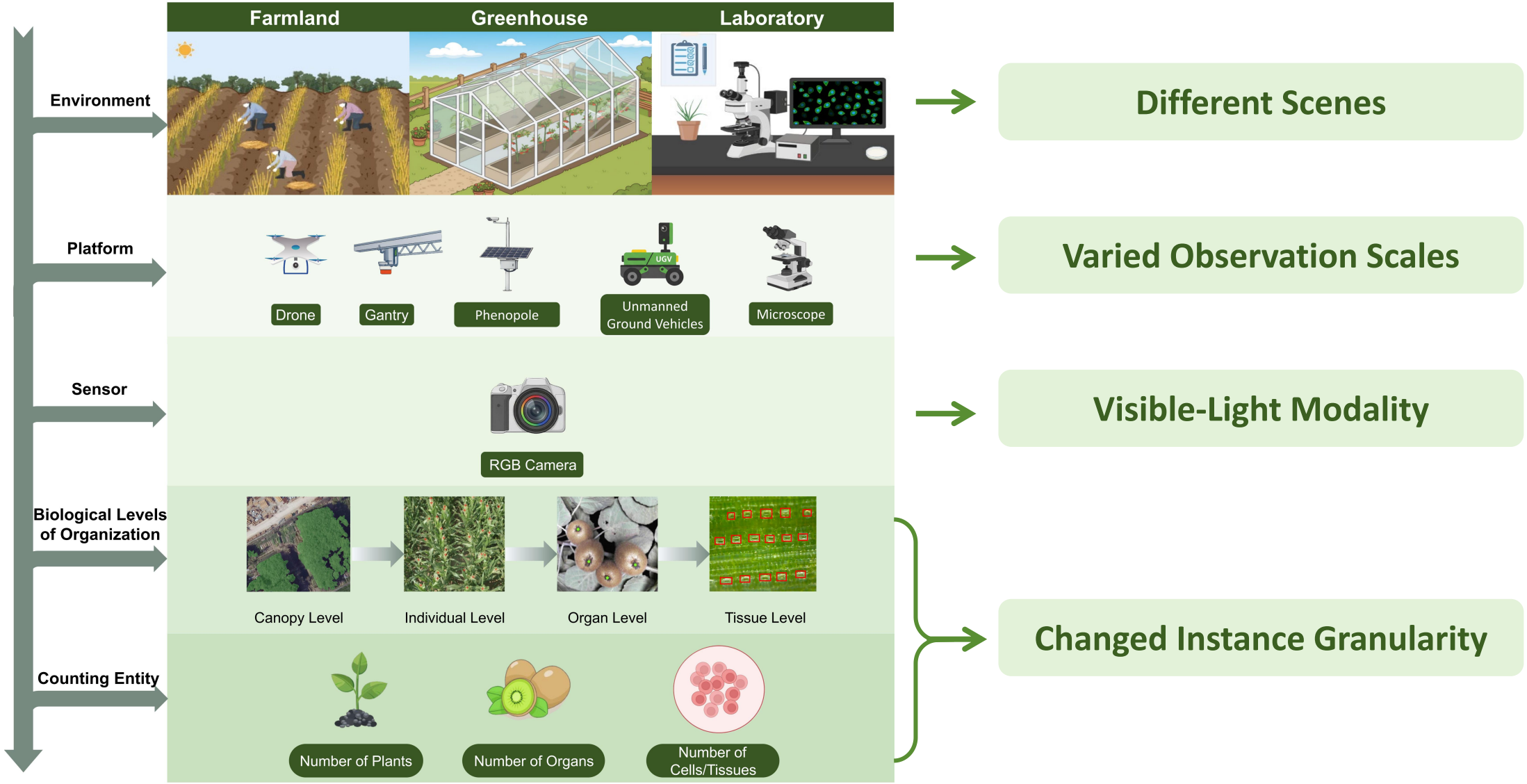
Taxonomy:
Plantae-Tracheophyta-Magnoliopsida-Sapindales-Burseraceae-Canarium-Album

Biological Organization:
Organ-Fruit

One image annotation contains points, boxes, taxonomy, and biological organization.

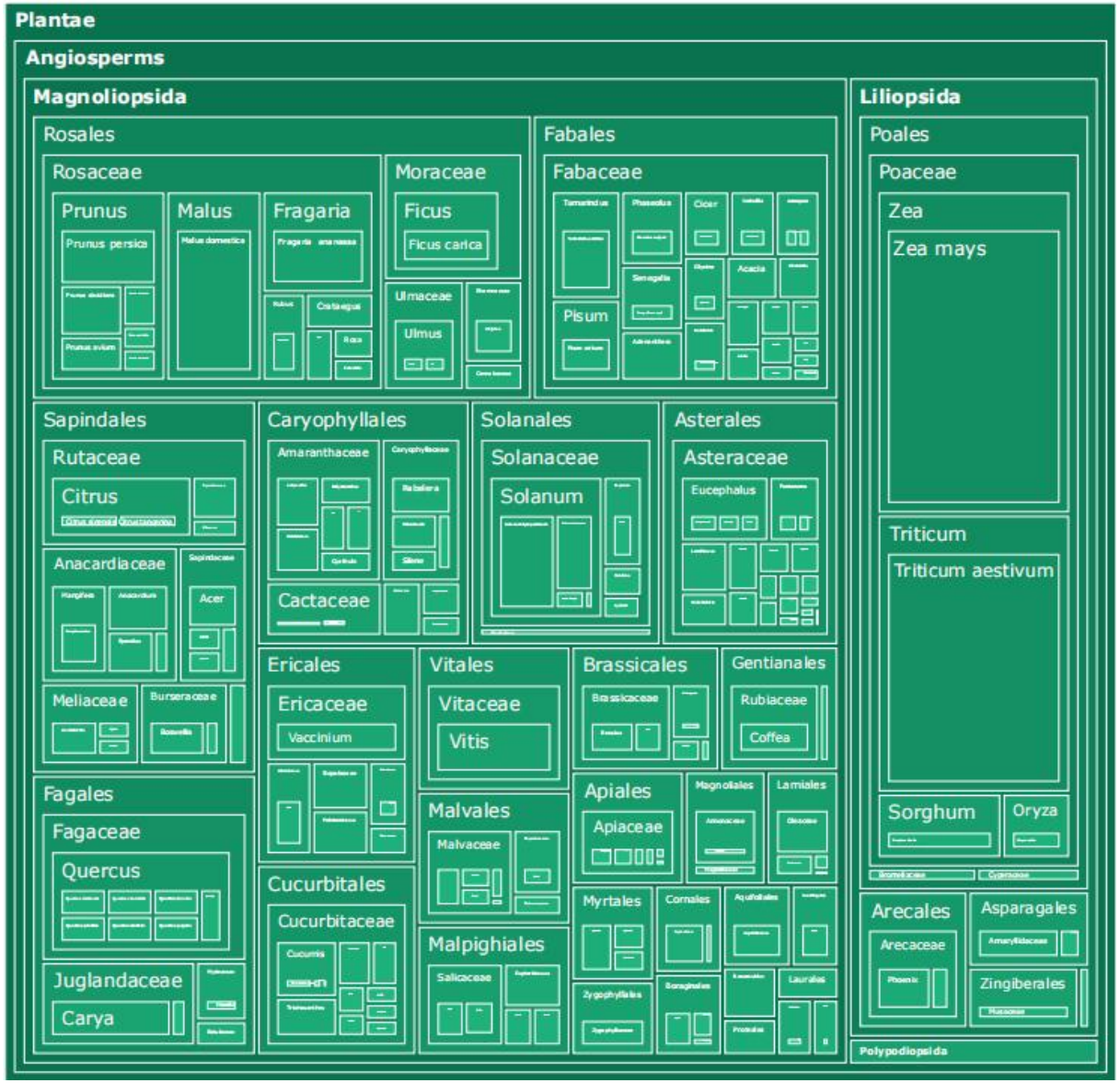
TPC–268 combines large-scale plant diversity with rich biological annotations.

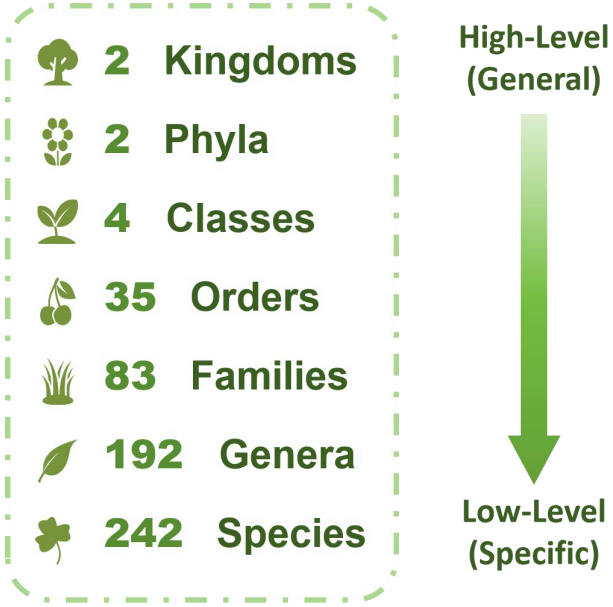
TPC-268 Spans Five Hierarchical Levels



Different levels introduce distinct visual challenges for plant counting.

A Taxonomic Map of Plant Diversity





TPC–268 organizes fine-grained **plant diversity** into a **biological hierarchy**, enabling taxonomy-aware counting and cross-species generalization.

Counting Across Biology and Scale

Category = Species × Biological Organization Example species: Maize (Zea mays)

The same species can look different fundamentally across organizations and scales.

Biological Organization axis

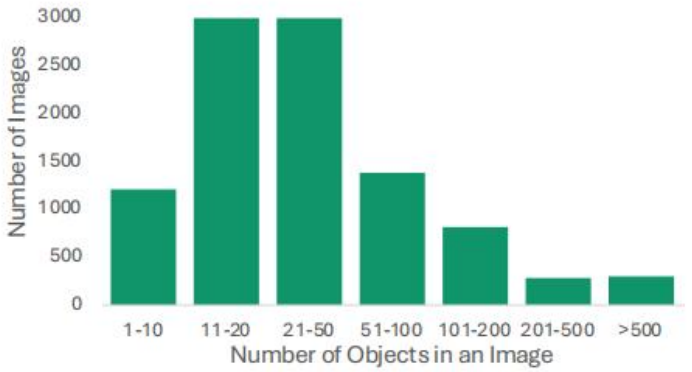
Tissue	Organ	Organism	Population
			
Microscope	Close-range Camera	Field Camera	Aerial View

Observation Scale axis

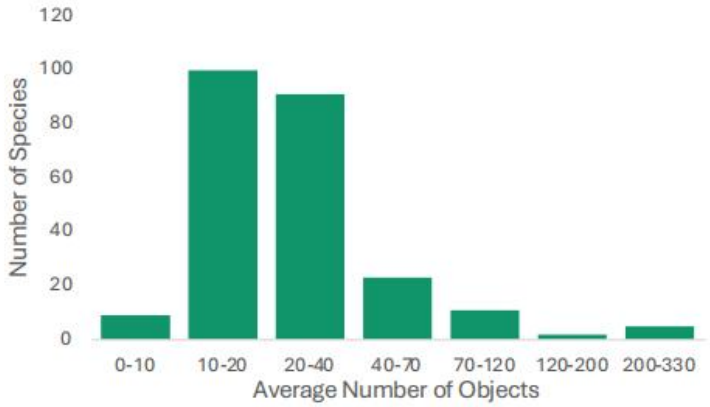
What to Count: Biologically-Grounded Plants

How to Observe: Micro-to-Macro Imaging

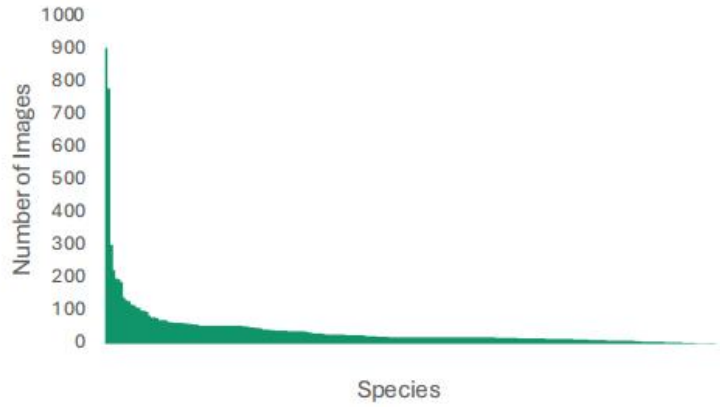
Real-World Plants Are Long-Tailed and Uneven



(b) Objects per image distribution



(c) Objects per species distribution



(d) Images per species from high to low

72.1% images contain <50 instances
but **3.0%** exceed 500 instances

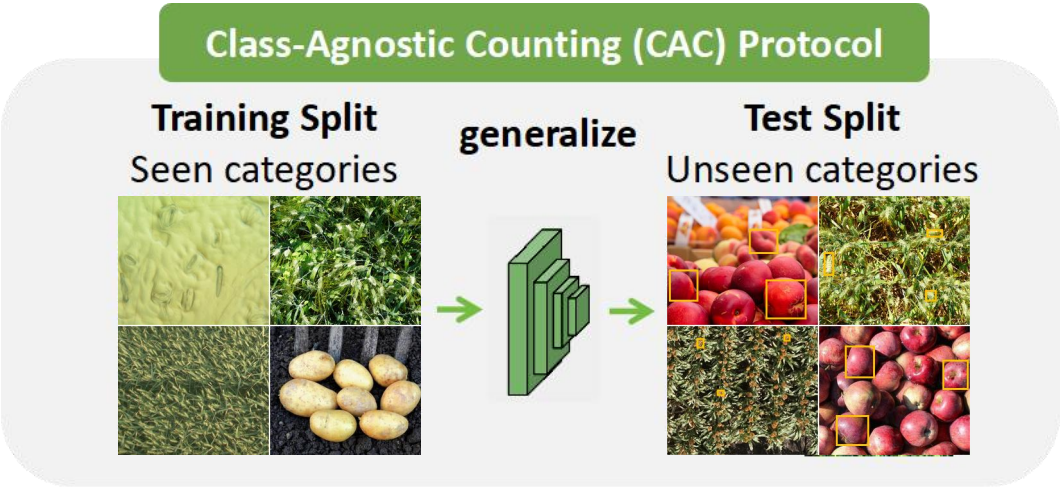
5–1,462 instances per image
from sparse samples to extremely dense scenes

53.5% species have <20 images
8.3% have ≤5 images

Top-10 Species cover **32.0%** of images
long-tail persists across taxonomic ranks

The benchmark captures both density variation and long-tailed biodiversity.

Current Models Are Not Ready for Plant Counting



Plant diversity motivates the CAC setting.

Train/test categories are strictly disjoint.

Test targets are specified by exemplar boxes.

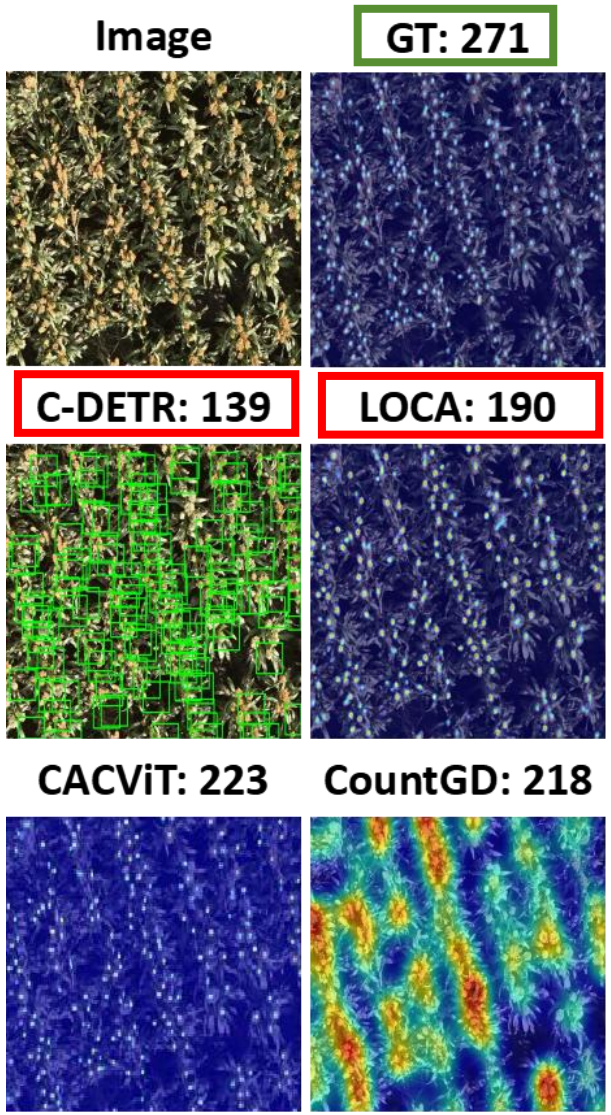
Table 2. **Comparison with the state-of-the-art CAC approaches on the TPC–268 dataset.** Best performance is in **boldface**.

Method	Venue & Year	Backbone	Shot	Val			Test		
				MAE↓	RMSE↓	R^2 ↑	MAE↓	RMSE↓	R^2 ↑
FamNet [43]	CVPR'21	R50	3	28.87	52.51	0.58	30.43	65.62	0.62
BMNet+ [45]	CVPR'22	R50	3	29.33	77.78	0.47	27.78	57.25	0.50
C-DETR [41]	ECCV'22	R50	3	22.66	77.51	0.75	22.68	57.97	0.74
SPDCNet [28]	BMVC'22	R18	3	25.66	72.49	0.52	23.70	47.53	0.64
CountTR [31]	BMVC'22	Hybrid	3	20.21	55.82	0.73	25.19	49.94	0.62
SAFECount [55]	WACV'23	R18	3	22.57	63.65	0.64	25.70	52.30	0.58
LOCA [49]	ICCV'23	R50	3	17.26	53.19	0.75	**17.51**	**38.37**	**0.78**
DAVE [42]	CVPR'24	R50	3	16.47	52.87	0.76	17.61	40.06	0.75
CACViT [53]	AAAI'24	ViT-B	3	16.63	**42.49**	0.82	22.04	41.79	0.73
CountGD [1]	NeurIPS'24	Swin-B	3	18.32	54.55	0.74	19.52	50.51	0.61
TasselNetV4 [21]	ISPRS'26	ViT-B	3	**13.20**	43.93	**0.83**	22.95	51.36	0.60
FamNet [43]	CVPR'21	R50	1	33.11±0.68	68.95±4.15	0.58±0.05	33.63±1.13	62.07±2.94	0.41±0.05
BMNet+ [45]	CVPR'22	R50	1	29.33±0.13	77.50±0.31	0.48±0.05	27.84±0.10	56.98±0.12	0.50±0.01
CountTR [31]	BMVC'22	Hybrid	1	20.16±0.05	55.15±0.82	0.73±0.01	25.19±0.14	50.23±0.24	0.62±0.00
LOCA [49]	ICCV'23	R50	1	17.19±0.31	48.14±2.19	0.80±0.02	21.47±0.29	**42.36**±0.72	**0.73**±0.01
DAVE [42]	CVPR'24	R50	1	16.06±0.60	48.35±1.19	0.80±0.01	**19.47**±0.44	42.54±0.35	0.72±0.00
CACViT [53]	AAAI'24	ViT-B	1	17.96±0.16	43.38±0.47	0.83±0.00	22.06±0.11	42.97±0.81	0.71±0.01
TasselNetV4 [21]	ISPRS'26	ViT-B	1	**13.49**±0.02	**41.30**±0.46	**0.85**±0.00	22.20±0.11	48.70±0.26	0.67±0.00

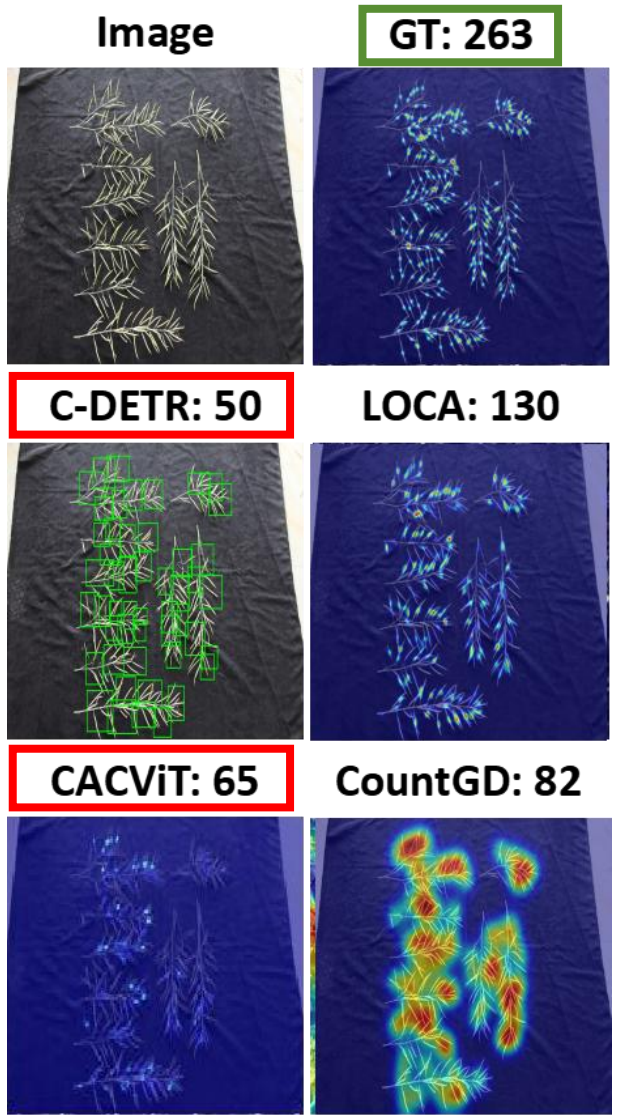
Best 3-shot result: LOCA 17.51 MAE / 38.37 RMSE

Even the best counting model leaves clear counting errors.

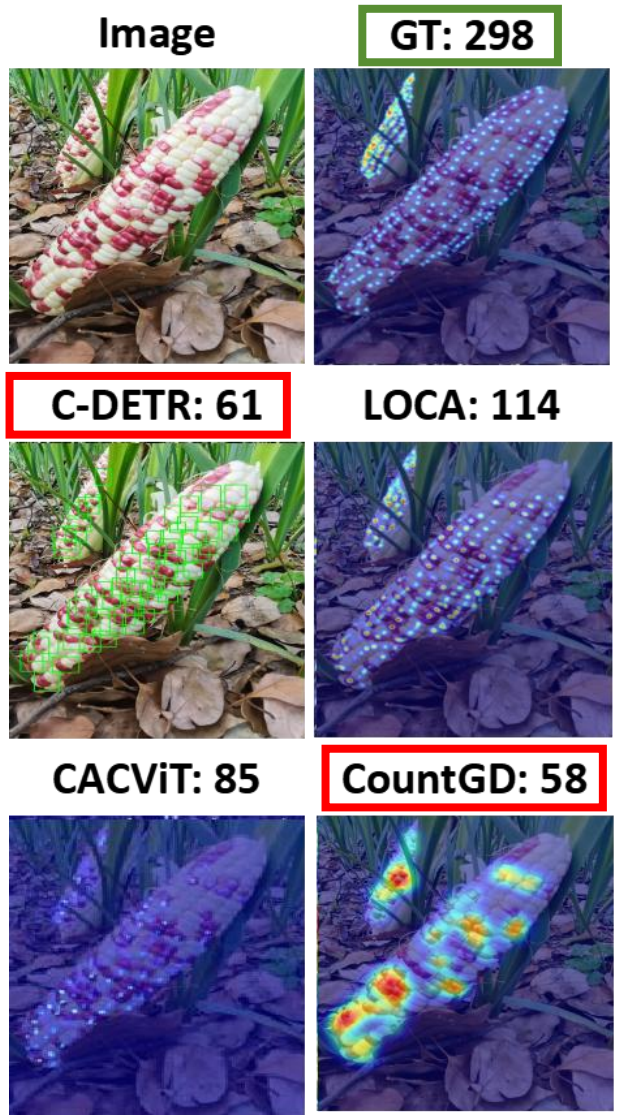
Seeing Why Models Struggle with Plants



Small-Scale Instances



Morphological Variations



Object Adhesion

Plant Counting Challenges Generic Counting

Table 3. **Results of cross-dataset transfer**. $A \rightarrow B$ denotes model trained on dataset A and tested on the test set of dataset B. Red/Blue indicate MAE increase/decrease compared to training and testing on the same dataset ($A \rightarrow A$).

Method	FSC–147→FSC–147			FSC–147→TPC–268			TPC–268→TPC–268			TPC–268→FSC–147		
	MAE	RMSE	R^2	MAE	RMSE	R^2	MAE	RMSE	R^2	MAE	RMSE	R^2
CountTR [31]	11.90	90.88	0.62	38.62 (+225%)	75.78	0.12	25.19	49.94	0.62	26.53 (+5%)	126.25	0.26
CACViT [53]	10.83	73.12	0.75	26.73 (+147%)	103.35	-0.63	22.04	41.79	0.73	17.88 (-19%)	82.57	0.68
LOCA [49]	10.72	56.10	0.85	24.70 (+130%)	68.68	0.59	17.51	38.37	0.78	15.16 (-13%)	109.15	0.45



FSC-147

Cross-Dataset Transfer



TPC-268

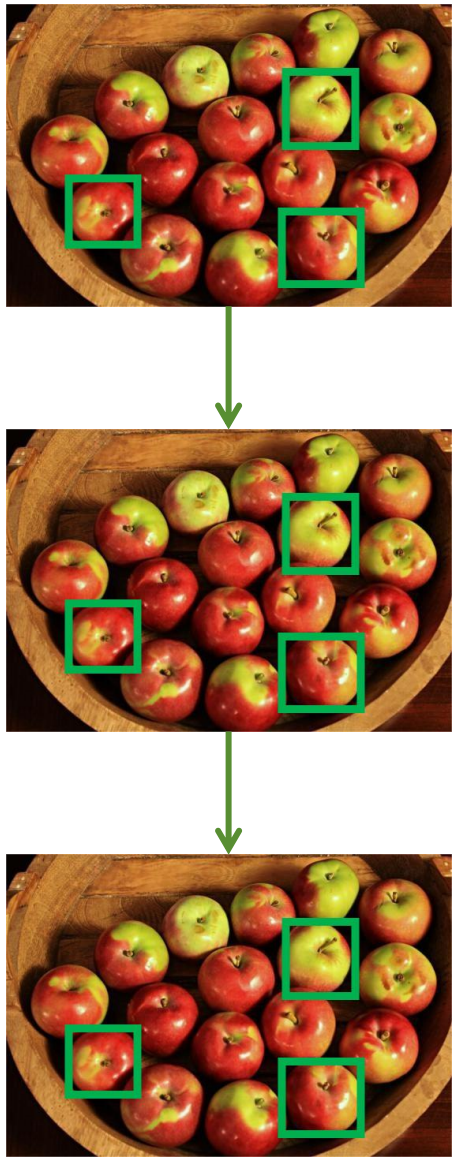
Models trained on FSC-147 show severe degradation on TPC-268, but not vice versa.

Taxonomy Helps, Biology Understanding Is Needed

Table 4. **Performance of CountGD on the TPC–268 test set with different prompts.** The species and taxonomic information are provided in text.

Target Specification	MAE↓	RMSE↓	R^2 ↑
3 visual exemplars	19.52	50.51	0.61
+ species name	17.53	44.80	0.69
+ full taxonomy	16.90	43.32	0.71

Taxonomy improves performance, even when used only as text prompts.



Visual: 3 exemplar boxes
Text: None
MAE: 19.52

Visual: 3 exemplar boxes
Text (species): Malus domestica
MAE: 17.53 **(-10.19%)**

Visual: 3 exemplar boxes
Text (full taxonomy):
Plantae, Tracheophyta, Magnoliopsida, Rosales, Rosaceae, Malus domestica
MAE: 16.90 **(-13.42%)**

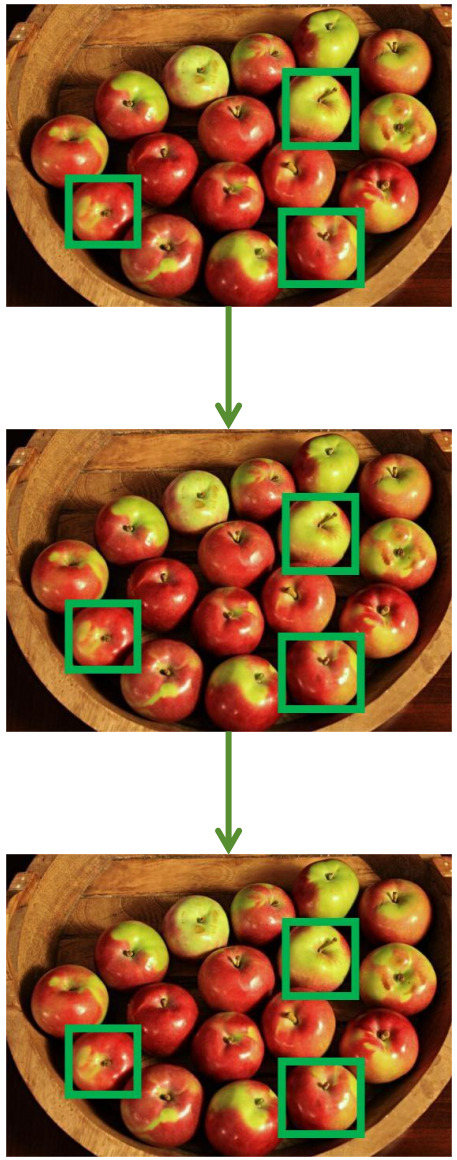
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Taxonomy improves performance, even when used only as text prompts.

? **How can we better use taxonomic hierarchy for plant counting?**

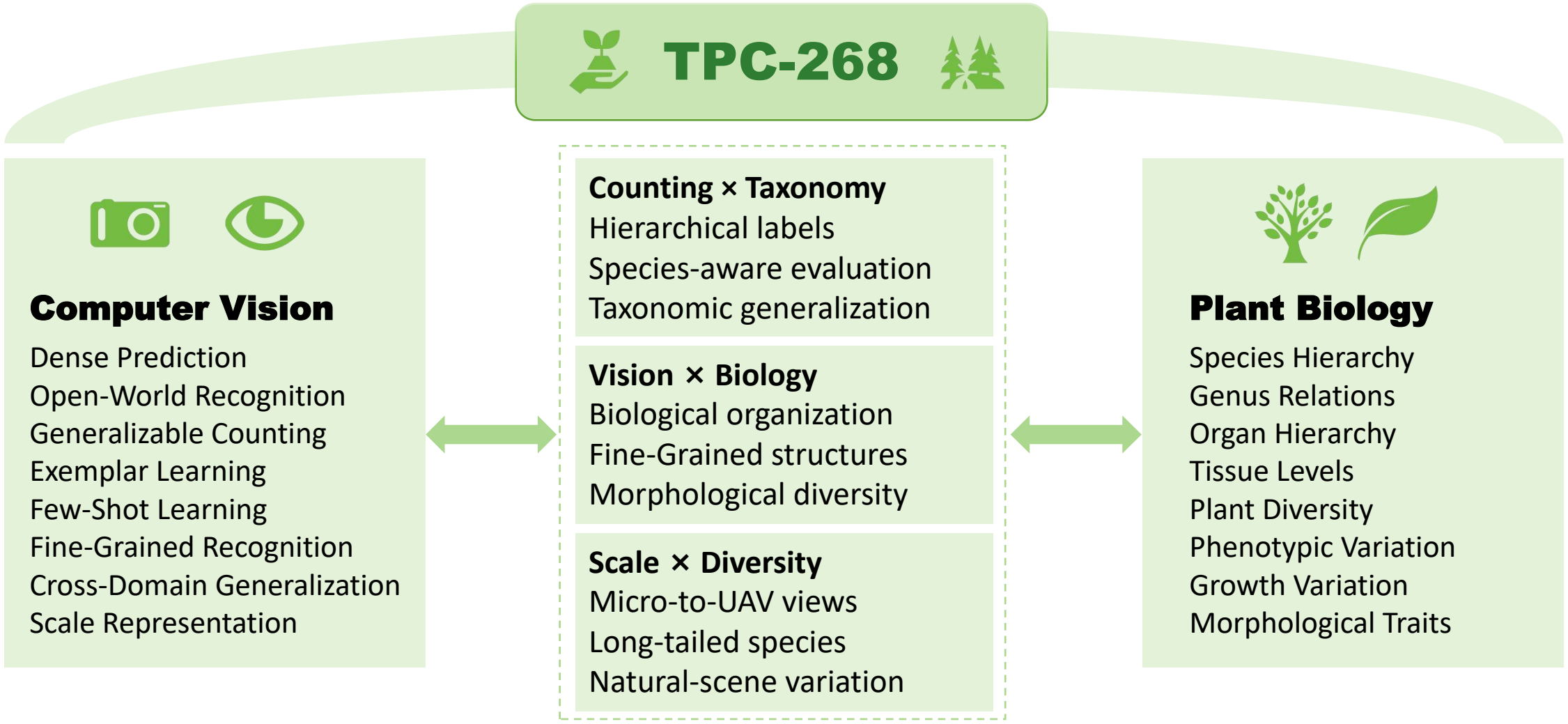


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Text (full taxonomy):
Plantae, Tracheophyta, Magnoliopsida, Rosales, Rosaceae, Malus domestica
MAE: 16.90 **(-13.42%)**

Bridging Computer Vision and Plant Biology

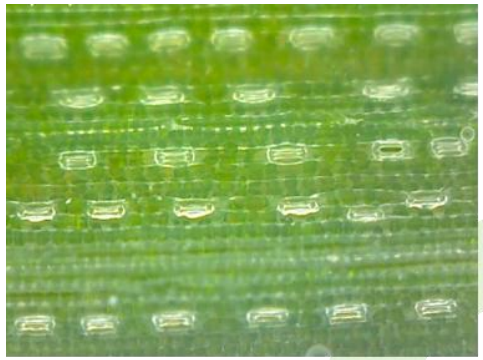


TPC–268 brings plant counting to a shared problem for vision and taxonomy.

From Counting Numbers to Understanding Nature

Beyond “how many”: what to count, where it sits in hierarchy, and at which biological level.

Cross-Scale Understanding
Micro-to-macro observation generalization







Cross-Level Targets
Counting across biological organizations

Taxonomic Hierarchy
Learning structured biological taxonomy







Biological Relatedness
Generalizing across related species

The next generation of plant counting models should reason over biological structure.

THANK YOU Q&A

Project page

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